```
    print(df.loc[sports, countries])
except KeyError as e:
    print(f"Error: One of the sports or countries you entered doesn't exist in the data.")
    print(f"Available sports: {df.index.unique().tolist()}")
```

```
Please enter sports(# as separator):  tennis
Please enter countries(# as separator):  usa
Error: One of the sports or countries you entered doesn't exist in the data.
Available sports: ['VOLLEYBALL', 'SOCCER', 'BASKETBALL', 'MARATHON', 'TENNIS', 'RUGBY', 'BOX', 'GOLF', 'CRICKET',
'F1', 'PING PONG', 'GYMNASTICS']
```

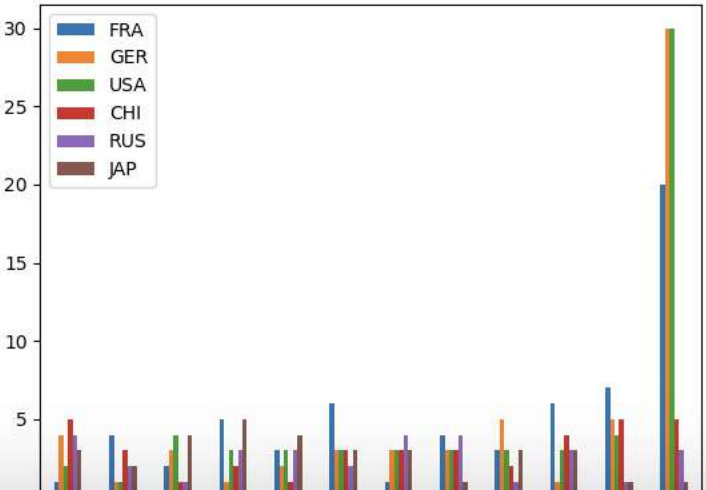
[2]:

```
import pandas as pd
import matplotlib.pyplot as plt
import wikipediaapi as wk

# read csv file
df = pd.read_csv("sports.csv")
df = df.set_index("SPORT")

# wikipedia object
wiki = wk.Wikipedia(
    language="en",
    user_agent="CIS2300-Project (student@example.com)"
)

print("CIS2300 Sports Project")

op = ""

while op != "7":
    print("\n*** Main Menu ***")
    print("1 - Display ALL data")
    print("2 - Display SELECTED records")
    print("3 - Add NEW sport")
    print("4 - Delete a sport")
    print("5 - Save to file")
    print("6 - Wikipedia info")
    print("7 - Exit")

    op = input("Please choose (1-7): ")

    # 1 show all data
    if op == "1":
        print("\n--- All Sports Data ---")
        print(df)
        g = input("Show graph? y/n: ")
        if g == "y" or g == "Y":
            df.plot(kind="bar")
            plt.xlabel("SPORT")
            plt.show()

    # 2 show selected data
    elif op == "2":
        print("\nCountries you can choose:", ", ".join(df.columns))
        countries = input("Enter countries (comma): ").upper().split(",")
        countries = [c.strip() for c in countries]

        print("Sports you can choose:", ", ".join(df.index))
        sports = input("Enter sports (comma): ").upper().split(",")
```

```python
# 2 show selected data
elif op == "2":
    print("\nCountries you can choose:", ", ".join(df.columns))
    countries = input("Enter countries (comma): ").upper().split(",")
    countries = [c.strip() for c in countries]

    print("Sports you can choose:", ", ".join(df.index))
    sports = input("Enter sports (comma): ").upper().split(",")
    sports = [s.strip() for s in sports]

    try:
        out = df.loc[sports, countries]
        print("\n--- Selected Data ---")
        print(out)
        g = input("Show graph? y/n: ")
        if g == "y" or g == "Y":
            out.plot(kind="bar")
            plt.xlabel("SPORT")
            plt.show()
    except:
        print("Something wrong. Maybe spelling is wrong.")

# 3 add new sport
elif op == "3":
    sp = input("Input sport name: ").upper()
    if sp == "":
        print("Sport name cannot empty.")
    else:
        vals = []
        for c in df.columns:
            v = input(c + " score: ")
            try:
                v = int(v)
            except:
                print("Not a number, set to 0.")
                v = 0
            vals.append(v)
        df.loc[sp] = vals
        print("New sport added.")

# 4 delete sport
elif op == "4":
    sp = input("Sport to delete: ").upper()
    if sp in df.index:
        df = df.drop(sp)
        print("Sport deleted.")
    else:
        print("Sport not exist.")

# 5 save file
elif op == "5":
    df.to_csv("sports.csv")
    print("File saved.")

# 6 wikipedia info
elif op == "6":
    kw = input("Search term: ")
    p = wiki.page(kw)
    if p.exists():
        print("\n--- Wikipedia Info ---")
        print(p.summary[:400])
```

```python
    # 6 wikipedia info
    elif op == "6":
        kw = input("Search term: ")
        p = wiki.page(kw)
        if p.exists():
            print("\n--- Wikipedia Info ---")
            print(p.summary[:400])
        else:
            print("No info found.")

    # 7 exit program
    elif op == "7":
        print("Ok, exit now.")

    # wrong option
    else:
        print("Wrong option, try again.")
```

```
CIS2300 Sports Project

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  1

--- All Sports Data ---
            FRA  GER  USA  CHI  RUS  JAP
SPORT
VOLLEYBALL    1    4    2    5    4    3
SOCCER        4    1    1    3    2    2
BASKETBALL    2    3    4    1    1    4
MARATHON      5    1    3    2    3    5
RUGBY         3    2    3    1    3    4
BOX           6    3    3    3    2    3
GOLF          1    3    3    3    4    3
CRICKET       4    3    3    3    4    1
F1            3    5    3    2    1    3
PING PONG     6    1    3    4    3    3
GYMNASTICS    7    5    4    5    1    1
D            20   30   30    5    3    1
Show graph? y/n:  y
```



Ruan_Jingwen_program.py | panda | Download file | iLovePDF

v4.20.0

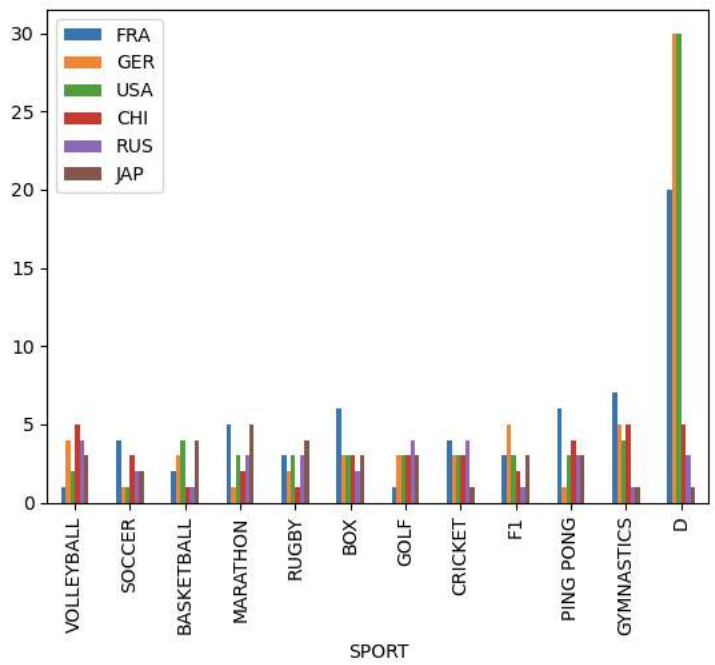
ojects

jupyter panda Last Checkpoint: yesterday

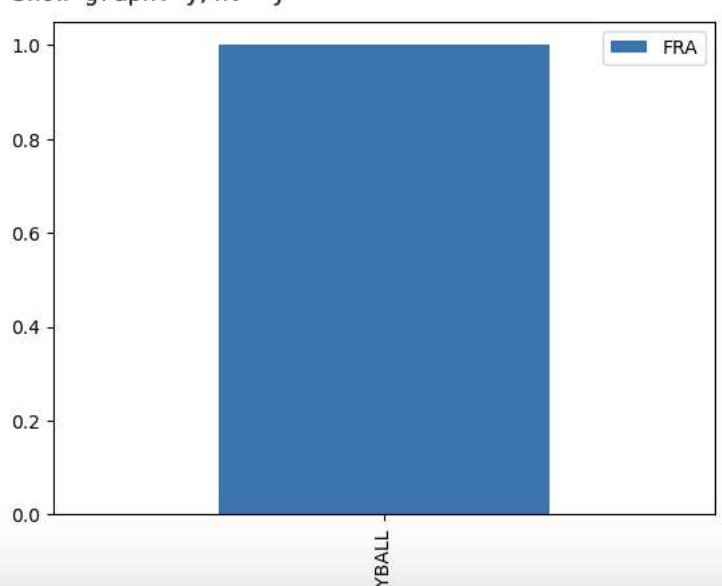
File Edit View Run Kernel Settings Help

Code | JupyterLab | Python [conda env:base] * | Trusted

```
Show graph? y/n:  y
```




```
*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  2

Countries you can choose: FRA, GER, USA, CHI, RUS, JAP
Enter countries (comma):  FRA
Sports you can choose: VOLLEYBALL, SOCCER, BASKETBALL, MARATHON, RUGBY, BOX, GOLF, CRICKET, F1, PING PONG, GYMNAST
ICS, D
Enter sports (comma):  VOLLEYBALL

--- Selected Data ---
            FRA
SPORT
VOLLEYBALL    1
Show graph? y/n:  y
```



localhost

Home | Ruan_Jingwen_program.py | panda | Download file | iLovePDF

Anaconda Toolbox v4.20.0

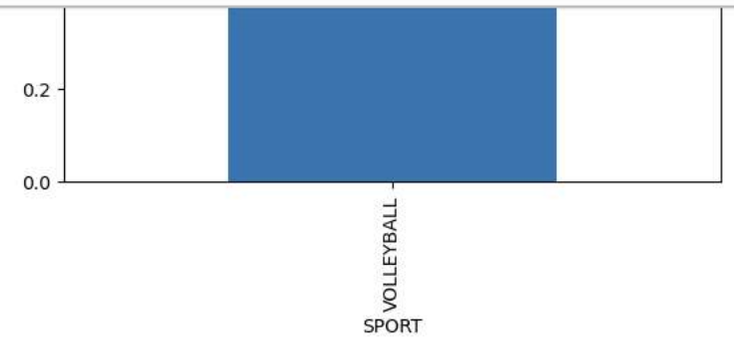
Anaconda

Your Toolbox for Python Projects

jupyter panda Last Checkpoint: yesterday

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *




```
*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  3
Input sport name:  ice skating
FRA score:  2
GER score:  1
USA score:  2
CHI score:  3
RUS score:  3
JAP score:  3
New sport added.

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  4
Sport to delete:  ice skating
Sport deleted.

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  5
File saved.

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  6
Search term:  china

--- Wikipedia Info ---
China, officially the People's Republic of China (PRC), is a country in East Asia. With a population exceeding 1
```

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] *

```
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  3
Input sport name:  ice skating
FRA score:  2
GER score:  1
USA score:  2
CHI score:  3
RUS score:  3
JAP score:  3
New sport added.

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  4
Sport to delete:  ice skating
Sport deleted.

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  5
File saved.

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  6
Search term:  china

--- Wikipedia Info ---
China, officially the People's Republic of China (PRC), is a country in East Asia. With a population exceeding 1.4
billion, it is the second-most populous country after India, representing 17% of the world population. China borde
rs fourteen countries by land across an area of 9.6 million square kilometers (3,700,000 sq mi), making it the thi
rd-largest country by area. The country is divided into 3

*** Main Menu ***
1 - Display ALL data
2 - Display SELECTED records
3 - Add NEW sport
4 - Delete a sport
5 - Save to file
6 - Wikipedia info
7 - Exit
Please choose (1-7):  7
Ok, exit now.
```